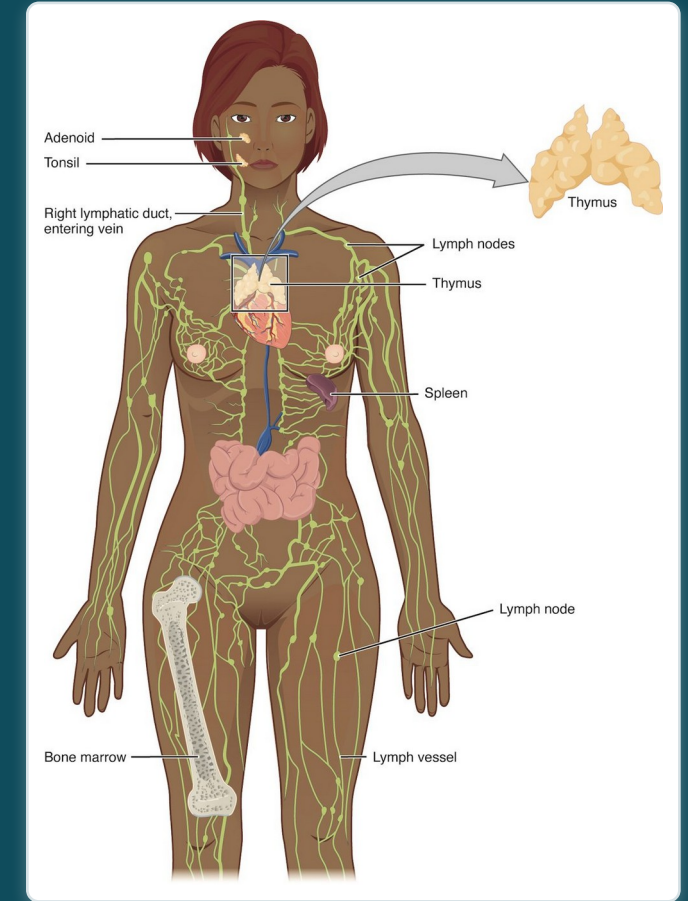


CHAPTER 21

The Lymphatic and Immune System

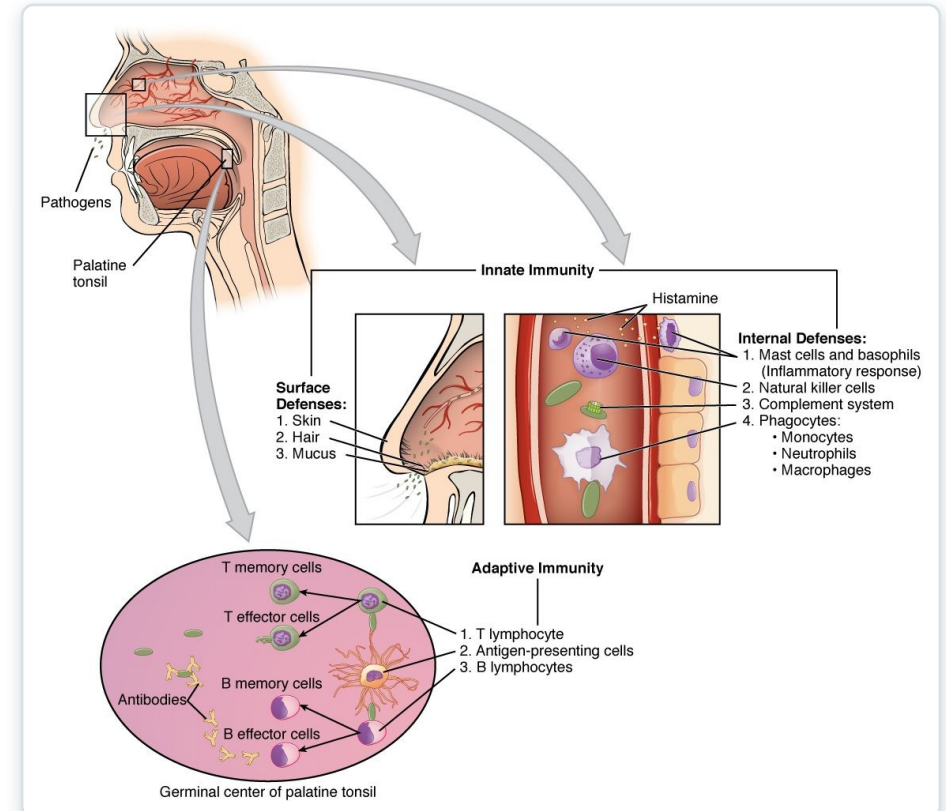
How the body defends itself against infection.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Identify the components and anatomy of the lymphatic system
- 2 Discuss the role of the innate immune response against pathogens
- 3 Describe the power of the adaptive immune response to cure disease
- 4 Explain immunological deficiencies and over-reactions of the immune system
- 5 Discuss the role of the immune response in transplantation and cancer



The body's innate and adaptive defenses.

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21.1

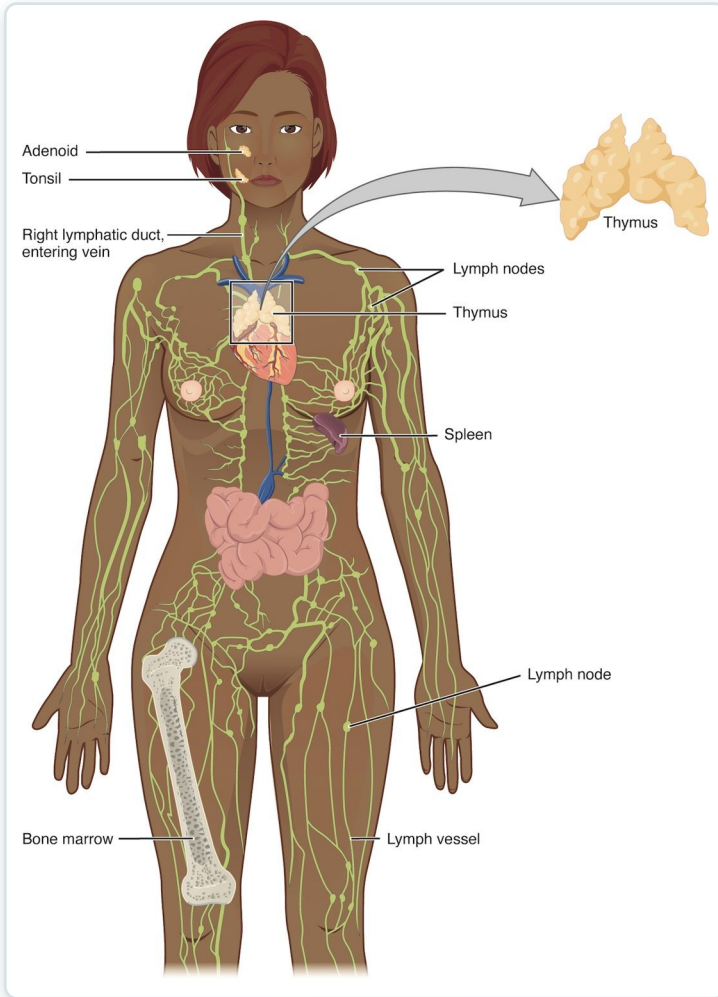
SECTION

Anatomy of the Lymphatic and Immune Systems

The lymphatic vessels and organs that drain fluid and house immune cells.

SECTION 21.1

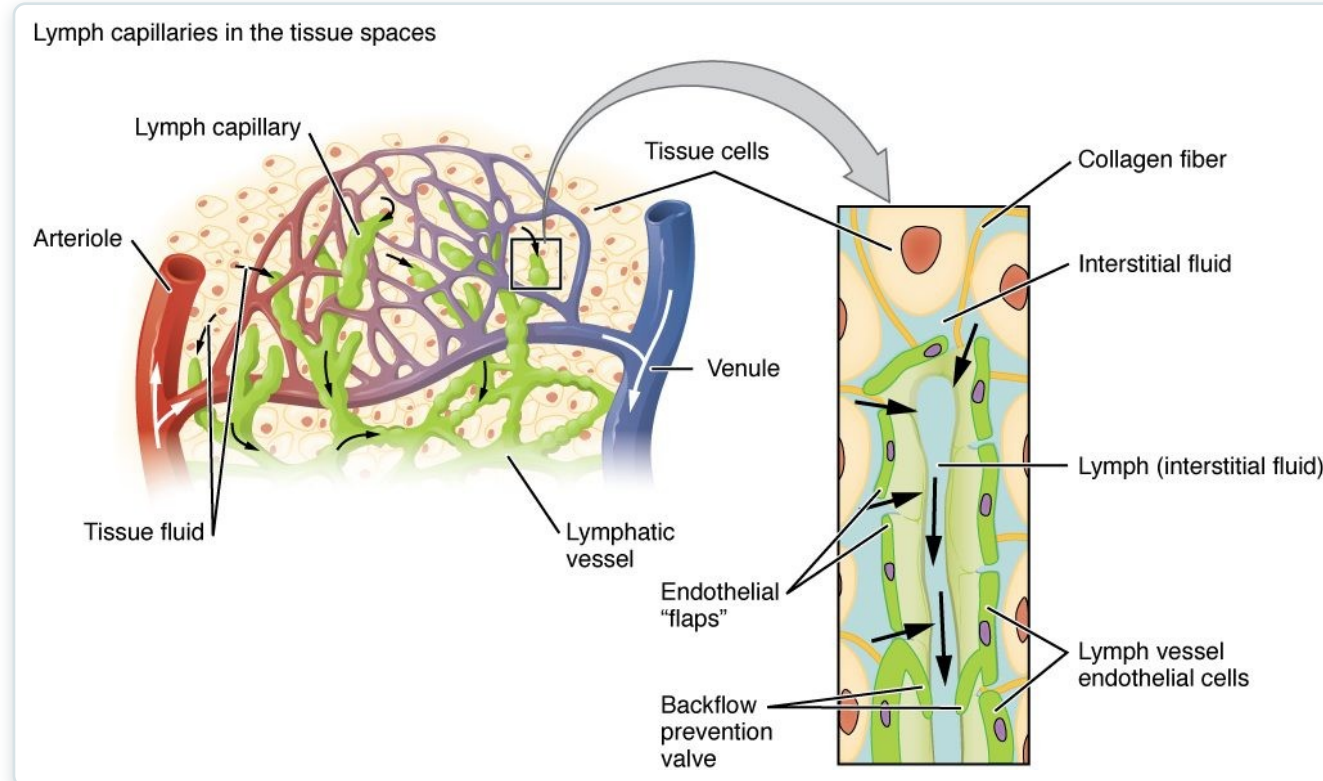
The Lymphatic System



The lymphatic system

- Vessels, nodes, and lymphoid organs
- Returns leaked fluid to the blood
- Houses and transports immune cells
- Includes spleen, thymus, tonsils

Formation of Lymph



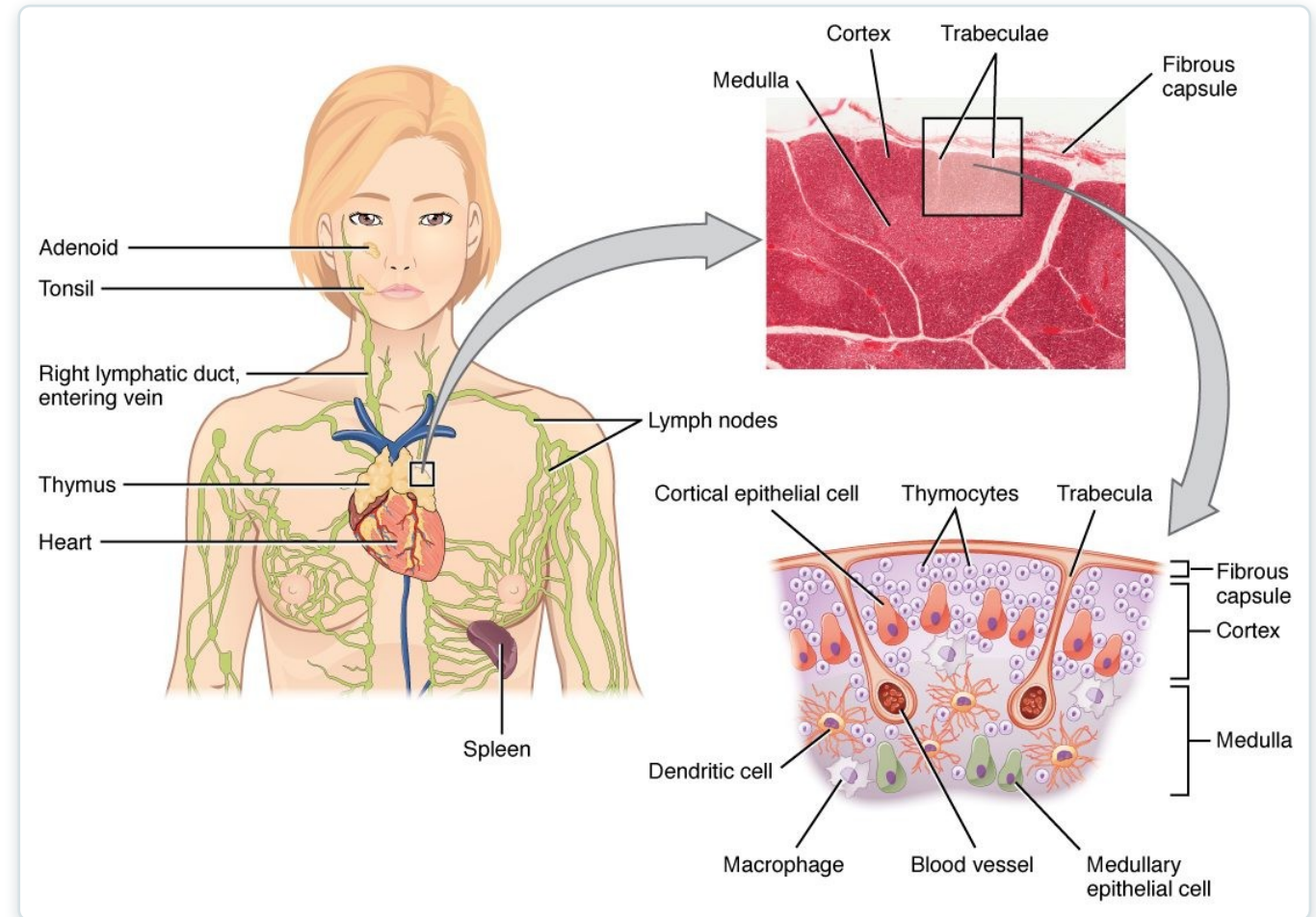
How lymph forms and drains

- Capillaries leak fluid into tissues
- Lymphatic capillaries collect it
- The fluid is now called lymph
- It returns to the bloodstream

SECTION 21.1

Lymph Nodes

- Small filtering organs along the vessels
- Trap pathogens and debris
- Packed with immune cells
- Swell during infection



The structure of a lymph node

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21.2

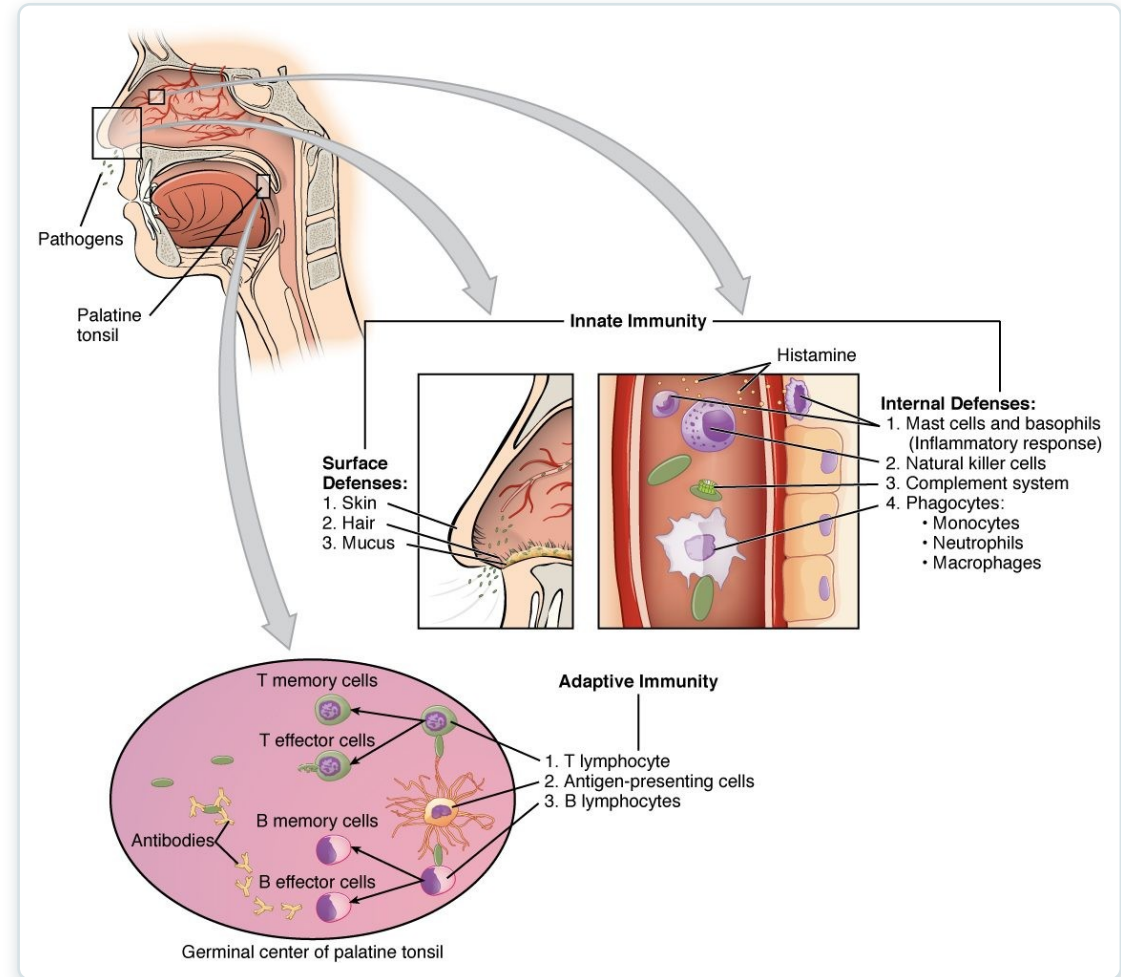
SECTION

Barrier Defenses and the Innate Immune Response

The body's first lines of defense — barriers and the innate response.

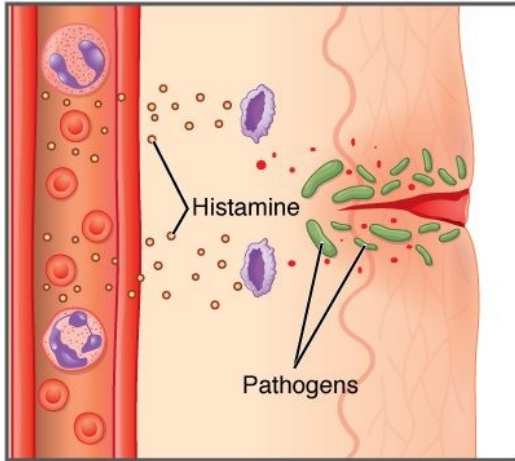
Innate vs. Adaptive Immunity

- Two arms of defense
- Innate: fast, general, present at birth
- Adaptive: slower, specific, learned
- Barriers are the first line

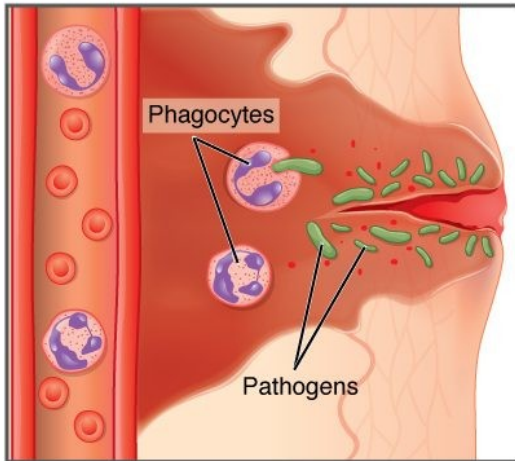


Innate and adaptive immunity

Inflammation



① Mast cells detect injury to nearby cells and release histamine, initiating inflammatory response.



② Histamine increases blood flow to the wound sites, bringing in phagocytes and other immune cells that neutralize pathogens. The blood influx causes the wound to swell, redden, and become warm and painful.

The inflammatory response

- A core innate response to injury
- Mast cells release histamine
- Vessels dilate and become leaky
- Causes redness, heat, swelling, pain

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SECTION

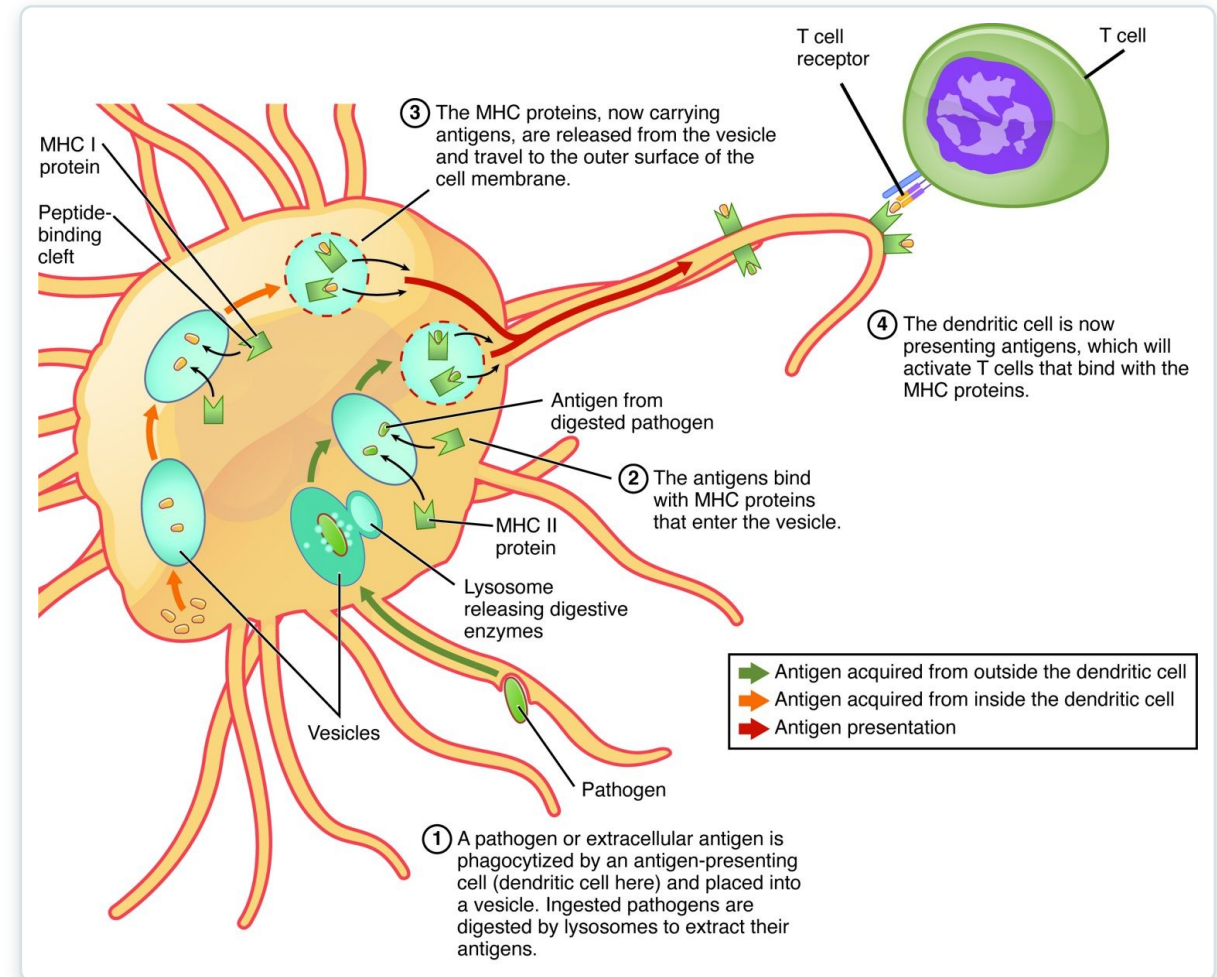
The Adaptive Immune Response: T lymphocytes and Their Functional Types

Adaptive immunity carried out by T lymphocytes.

SECTION 21.3

Antigen Presentation

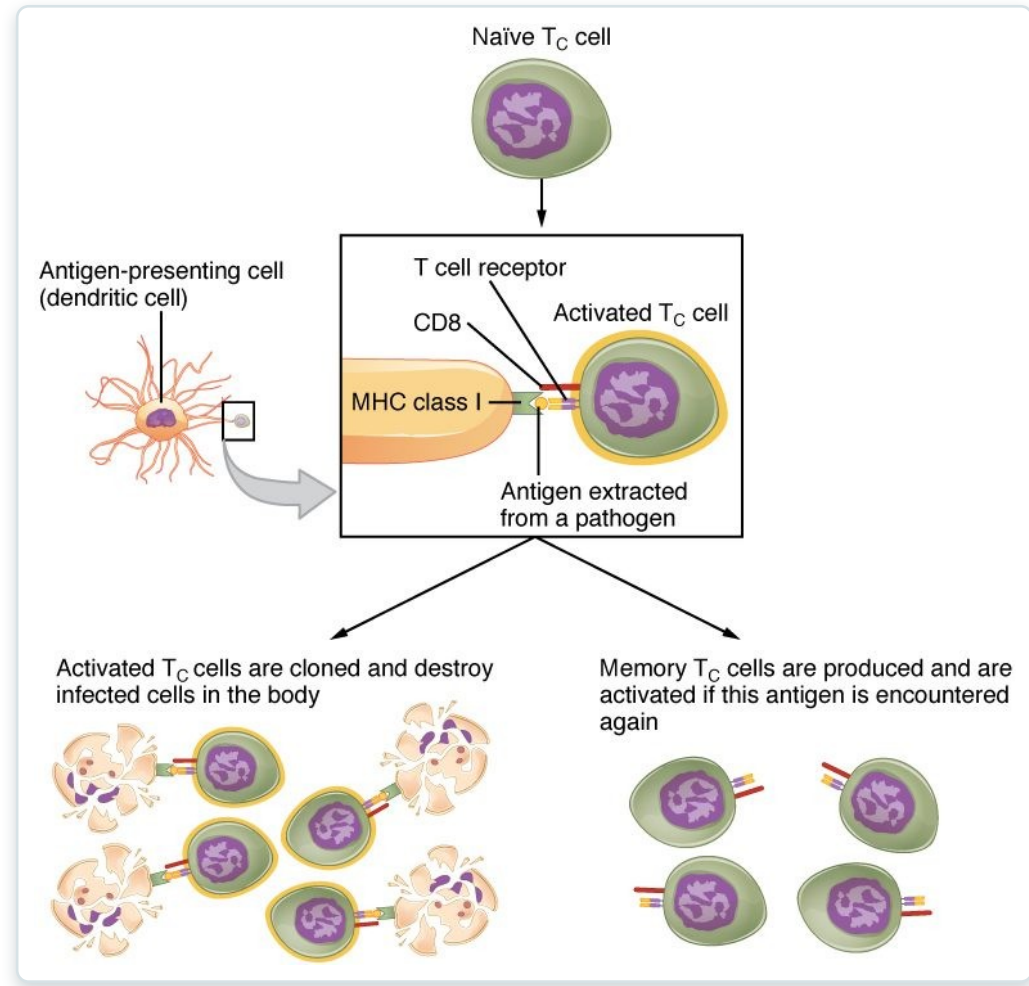
- T cells must be shown the enemy
- Antigen-presenting cells display antigens
- Displayed on MHC molecules
- This activates the matching T cell



Antigen presentation to a T cell

Types of T Cells

- Helper T cells coordinate the response
- Cytotoxic T cells kill infected cells
- Memory T cells remember the pathogen
- Central to adaptive immunity



Helper and cytotoxic T cells

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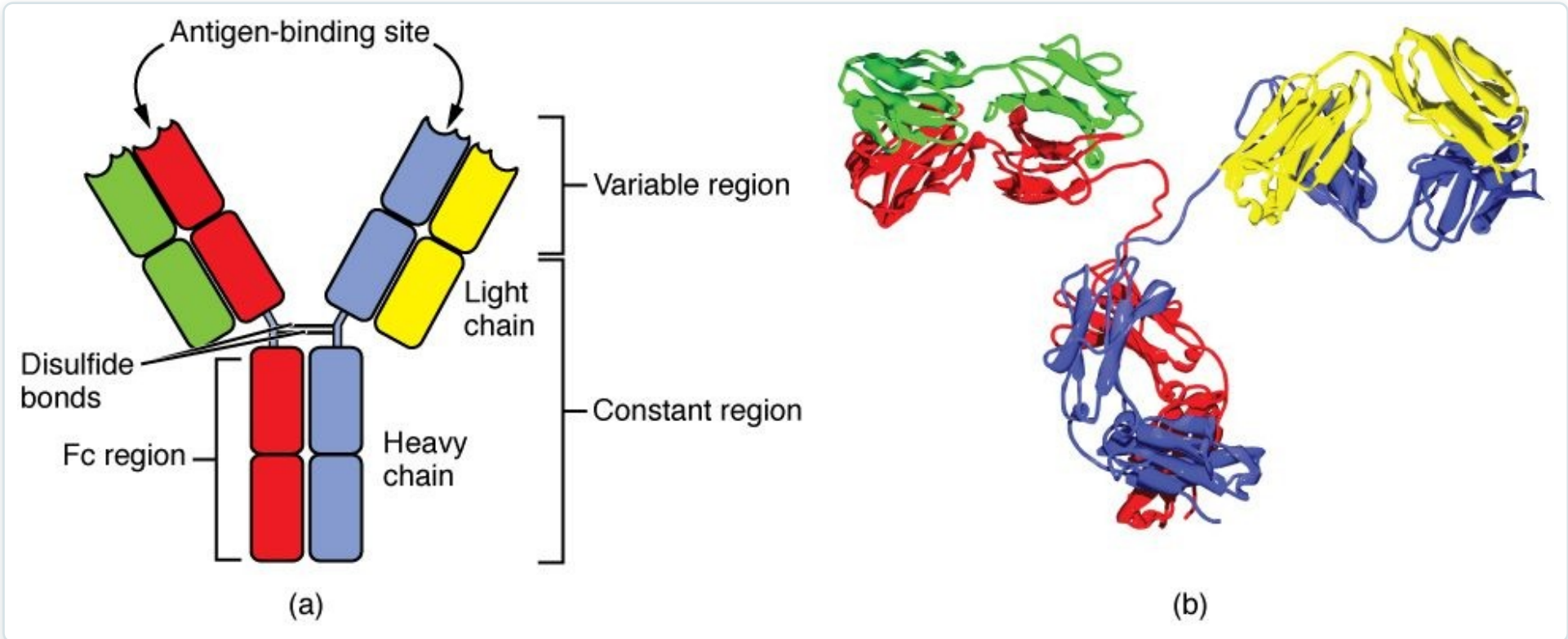
21.4

SECTION

The Adaptive Immune Response: B-lymphocytes and Antibodies

Adaptive immunity carried out by B lymphocytes and antibodies.

Antibody Structure

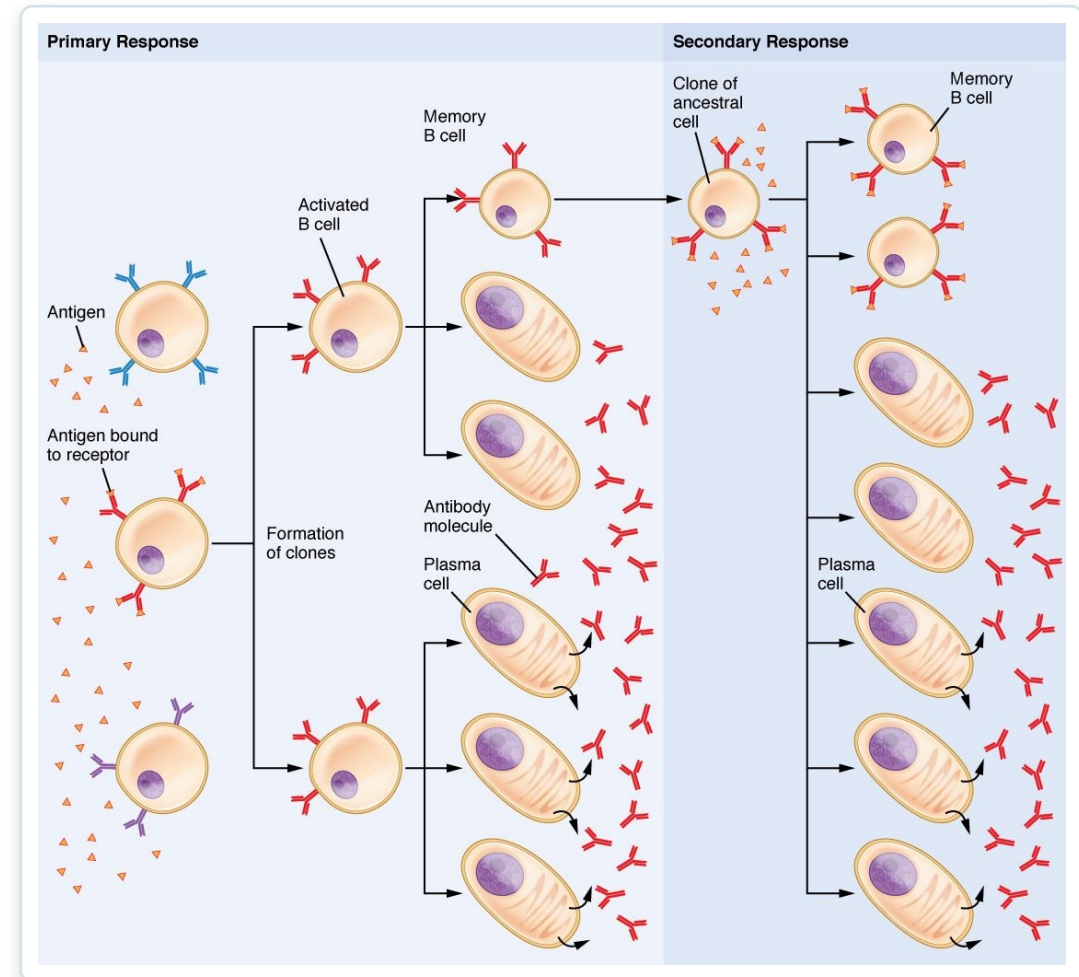


The structure of an antibody

- Y-shaped proteins made by B cells
- Two antigen-binding tips
- Each binds one specific antigen
- Tag pathogens for destruction

B Cell Activation

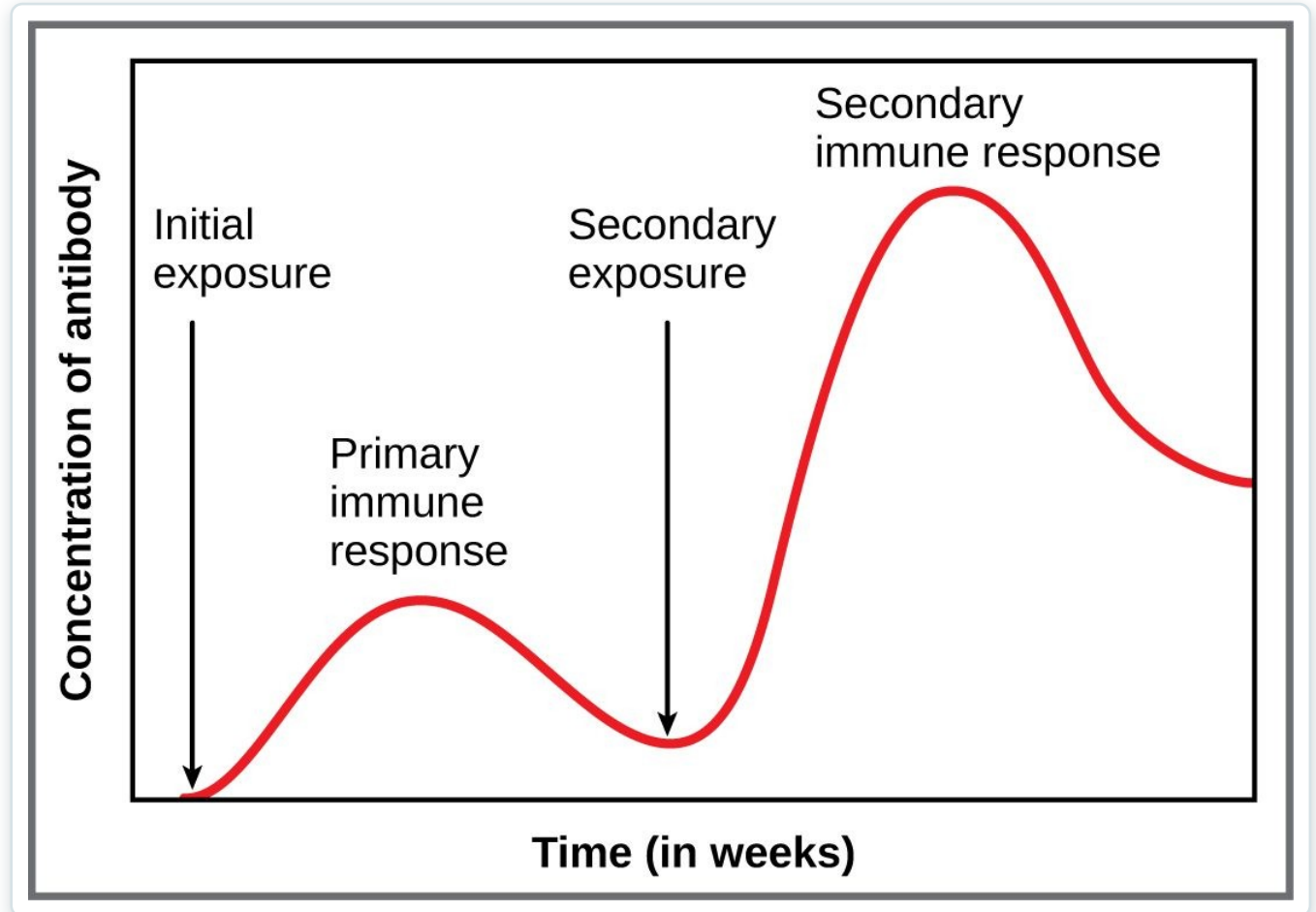
- B cells make antibodies
- Activated B cells become plasma cells
- Plasma cells pump out antibodies
- Memory B cells remain behind



B cell activation and antibody production

Primary vs. Secondary Response

- First exposure → slow, weak response
- Memory cells are formed
- Second exposure → fast, strong response
- The principle behind vaccines



The primary and secondary immune response



21.5

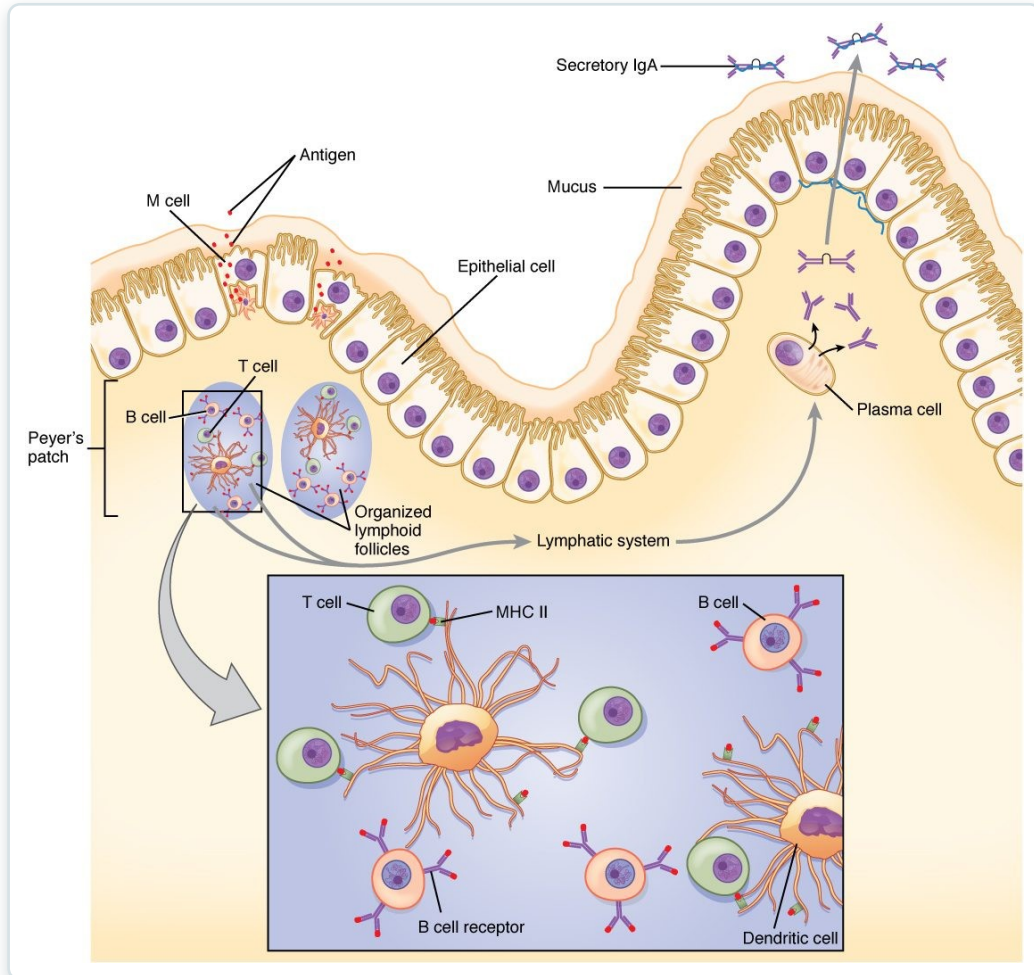
SECTION

The Immune Response against Pathogens

How the immune defenses work together against an infection.

A Coordinated Immune Response

- The defenses work as a team
- Barriers and innate cells respond first
- Adaptive immunity targets the pathogen
- Memory protects against reinfection



A coordinated immune response

Key Takeaways

- The lymphatic system drains tissue fluid and houses the immune system.
- Innate immunity provides fast, general defense; adaptive immunity is specific and remembers.
- Inflammation brings defenses to the site of injury or infection.
- T cells coordinate the response and kill infected cells; B cells make antibodies.
- Immune memory underlies long-term protection and the success of vaccines.

CLINICAL CONNECTIONS

Why the Immune System Matters in Practical Nursing

Vaccination

Builds immune memory without causing the disease.

Allergies

An overactive immune response to harmless substances.

Autoimmune Disease

The immune system attacks the body's own tissues.

HIV/AIDS

The virus destroys helper T cells, crippling immunity.

Lymph Node Exam

Swollen nodes can signal infection or, sometimes, cancer.

Immunocompromise

Weakened immunity raises infection risk in many patients.